

Qatar-2b

R-1279

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160319 · created 2026-07-31T14:43:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457466.81338 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.065
Transit depth (Rp/Rs)^2	3.85 +/- 2.57 [%]
Orbital Inclination (inc)	85.24 +/- 1.7
Airmass coefficient 1 (a1)	0.59 +/- 0.31
Airmass coefficient 2 (a2)	0.208 +/- 0.076
Scatter in the residuals of the lightcurve fit is	50.31 %
Best Comparison Star	#3 - [270, 135]
Optimal Aperture	2.53
Optimal Annulus	7.0
Transit Duration (day)	0.0698 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.065 vs 0.16327 (deviation 0.38 σ)
Duration fit vs published	0.0698 d vs 0.0767 d (deviation 0.57 σ)
Mid-time O-C	-0.5 min
Depth SNR	2.3
Residual scatter	50.31 %

Light curve

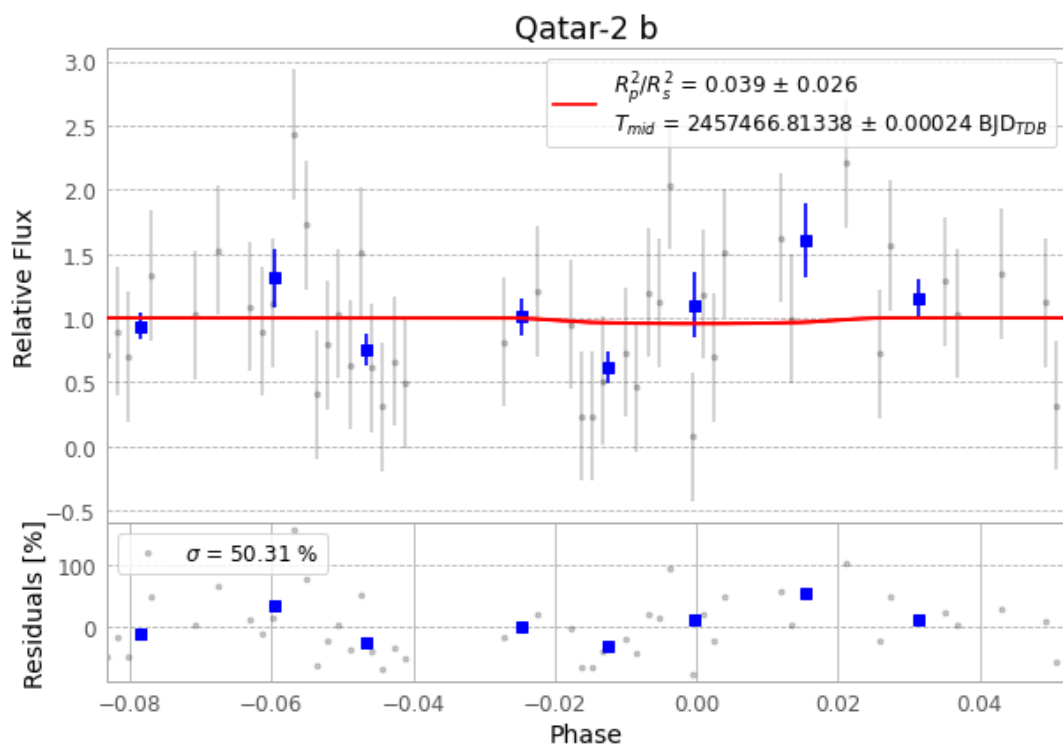


Figure 1 — FinalLightCurve_Qatar-2 b_2016-03-18.png (SHA-256 78faf25ed1579fc6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [197, 367], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 274ca258d805d3c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), Qatar-2160319044212.FITS → Qatar-2160319090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160319.log (077db41d6c01...), FinalLightCurve_Qatar-2 b_2016-03-18.png (78faf25ed157...), FinalLightCurve_Qatar-2 b_2016-03-18.csv (3e6c4d16e703...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:43Z.